

EQUAZIONE CONGRUENZIALE E DIOFANTEA

$$5x \equiv_3 1 \quad (\text{Mod } 3)$$

e.d.

$$5x + 3y = 1$$

$$a = 5$$

$$b = 3$$

$$c = 1$$

$$1) \quad 5 = 3 \cdot 1 + 2$$

$$3 = 2 \cdot 1 + 1$$

$$2 = 1 \cdot 1 + 1 \rightarrow \text{MCD}(5, 3) = 1 = d$$

$$1 = 1 \cdot 1 + 0$$

2) $d | c$? $1 | 1$ Sì, 7 una soluzione di e.d.

3) IDENTITÀ di BEZOUT

$$\begin{cases} 5 = 3 \cdot 1 + 2 \\ 3 = 2 \cdot 1 + 1 \\ 2 = 1 \cdot 1 + 1 \end{cases} \rightarrow \begin{cases} z = 5 - 3 \cdot 1 \\ 1 = 3 - 2 \cdot 1 \\ \underline{1 = 2 - 1 \cdot 1} \end{cases}$$

$$\begin{aligned} 1 &= 2 - 1 \cdot 1 = 2 - (3 - 2 \cdot 1) = 5 - 3 \cdot 1 - (3 - (5 - 3 \cdot 1)) \\ &= 5 - 3 - (3 - 5 + 3) = 5 - 3 - 3 + 5 - 3 = \underset{11}{5}(\underset{7}{2}) + \underset{11}{3}(\underset{5}{-3}) \end{aligned}$$

$$c = d \cdot \tau$$

$$5 = 3 \cdot 2 + (-1) \quad 5 \equiv_3 -1$$

$$-x \equiv_3 1$$

$$x \equiv_3 -1 = [x]_3$$

$$x = -1 = 3 \cdot 1 - 4 = -1$$

check

$$-20 \equiv_3 1 \rightarrow 3 \mid -20 - 1 \rightarrow 3 \mid -21 \checkmark$$

$$-5 \equiv_3 1 \rightarrow 3 \mid -6 \checkmark$$

$$1 = 1 \cdot 1$$

$$1 = [5(2) + 3(-3)] \cdot 1$$

$$1 = 5(2) + 3(-3)$$

$$(x', y') = (2, -3)$$

$$x = 2$$

$$5x \equiv_3 1$$

$$3 \mid 10 - 1 = 3 \mid 9 \quad \checkmark$$

$$X = -1$$

$$X = 2$$

FANNO PARTE DELLA STESSA CLASSE?

$$X = -1$$

$$2 = 3 \cdot 1 - 1 = -1$$



